```
# -*- coding: utf-8 -*-
"""Untitled11.ipynb

Automatically generated by Colab.

Original file is located at
    https://colab.research.google.com/drive/1YnP6aS3KR0P9miiyV4UzzFfoJeqa79Ym
"""

!pip install -q polars gdown

import polars as pl

!gdown 1iYqAGTtIhRLK4ycFK41hYaWK3iKfOdT2 -O title.basics0.tsv.gz
!gdown 1kZuj1lnXkPrNURzwYuc4FXTM7Pvknr3o -O title.ratings.tsv.gz
!gdown 1oLR2_mFyRHEiKlqFKt4QDVYlGH8LC4mn -O title.principals0.tsv.gz

basics = pl.scan_csv(
    "title.basics0.tsv.gz",
    separator="\t",
    null_values="\\N")

ratings = pl.scan_csv(
    "title.ratings.tsv.gz",
    separator="\t",
    null_values="\\N")

principals = pl.scan_csv(
    "title.principals0.tsv.gz",
    separator="\t",
    null_values="\\N")

print(basics.collect_schema())
print(ratings.collect_schema())
print(principals.collect_schema())

basics.head(5).collect()

ratings = ratings.with_columns(
    pl.col("averageRating").cast(pl.Float64),
    pl.col("numVotes").cast(pl.Int64))

filmes = (basics
    .filter(pl.col("titleType") == "movie")
    .join(ratings, on="tconst", how="inner"))
filmes.head(5).collect()

# ex1

resultado1 = ( filmes
    .select(
        "primaryTitle",
        "averageRating",
        "numVotes"
    )
    .sort(
        ["averageRating", "numVotes"],
        descending=[True, True]
    )
    .head(5)
    .collect())

resultado1

#ex2

resultado2 = (
    filmes
    .filter(pl.col("averageRating") > 8)
    .filter(pl.col("genres").is_not_null())
```

```
    .with_columns(
        pl.col("genres").str.split(",")
    )
    .explode("genres")
    .group_by("genres")
    .len()
    .sort("len", descending=True)
    .head(1)
    .collect())

resultado2

#ex3

resultado3 = ( principals
    .filter(
        pl.col("category").is_in(["actor", "actress"])
    )
    .join(
        filmes
        .filter(pl.col("averageRating") > 7.5)
        .select("tconst"),
        on="tconst",
        how="inner"
    )
    .select("nconst", "tconst")
    .unique()
    .group_by("nconst")
    .len()
    .sort("len", descending=True)
    .head(3)
    .collect())

resultado3

from datetime import datetime
from zoneinfo import ZoneInfo

# Define o fuso horário de Brasília
fuso_brasilia = ZoneInfo("America/Sao_Paulo")

# Pega o horário atual na zona especificada
horario_brasilia = datetime.now(fuso_brasilia)

print("Horário de Brasília:", horario_brasilia.strftime("%d/%m/%Y %H:%M:%S"))
```